Day 5
Git & GitHub

Version control • Repositories • Commits • Branching • README

Task: Publish the Day 1-4 internship programs on GitHub

1. Concepts

- **Git** is a version control system installed on your computer. It records snapshots (commits) of your project so you can track changes and go back to earlier versions.
- **GitHub** is a website that hosts Git repositories online, so your work is backed up, shareable and visible to others.
- **Repository (repo)** is a project folder tracked by Git.
- **Commit** is a saved snapshot with a message. **Branch** is an independent line of work. **Remote** is the online copy (usually named `origin`).

The three areas of Git

Area	Meaning	Command that moves files here
Working directory	Your files as you edit them	(you edit files)
Staging area	Changes chosen for the next commit	git add
Local repository	Committed history on your computer	git commit
Remote (GitHub)	Online copy of the repository	git push

2. Setup

Step 1: Create a GitHub account

- Go to github.com and click Sign up. Choose a professional username, as it appears in your profile link.
- Verify your email address.

Step 2: Install Git

- **Windows:** download from git-scm.com and install with the default options (this includes Git Bash).
- **macOS:** run `brew install git` or install the Xcode command line tools.
- **Linux (Ubuntu):** run `sudo apt install git`.

```
git --version
```

Step 3: Configure your identity (one time)

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global init.defaultBranch main
git config --list
```

Use the same email as your GitHub account so commits are linked to your profile.

3. Create the Repository on GitHub

- Click the + icon (top right) and choose New repository.
- Name: `java-internship-tasks`.
- Description: "Programs and exercises from my Java internship".
- Choose Public (or Private if your mentor prefers).
- Tick "Add a README file" only if you plan to clone first. If you will push an existing folder, leave it unticked.

• Click Create repository.

4. Prepare Your Local Folder

Arrange your programs like this:

```
java-internship-tasks/
  Day1/        (your Day 1 programs)
  Day2/        (your Day 2 programs)
  Day3/        (your Day 3 programs)
  Day4/
    StudentManagement.java
  README.md
  .gitignore
```

Create a `.gitignore` so compiled and generated files are not uploaded:

```
*.class
students.csv
.idea/
*.iml
out/
.vscode/
```

5. Core Workflow: init, add, commit, push

```
cd java-internship-tasks
git init                          # start tracking this folder
git status                        # see what Git notices
git add .                         # stage everything
git commit -m "Add Day 1-4 programs and README"
git branch -M main                # make sure branch is called main
git remote add origin https://github.com/<username>/java-internship-tasks.git
git push -u origin main           # first upload
```

Authentication

GitHub no longer accepts your account password for pushes over HTTPS. When Git asks you to sign in, use one of these:

- A browser sign-in window (Git Credential Manager, included with Git for Windows and recent macOS installs).
- A **Personal Access Token**: GitHub → Settings → Developer settings → Personal access tokens. Paste the token where the password is requested.
- An **SSH key**, or the GitHub CLI: `gh auth login`.

6. Command Reference

Command	What it does
git init	Creates a new repository in the current folder
git clone <url>	Downloads a repository from GitHub to your computer
git status	Shows changed, staged and untracked files
git add <file> / git add .	Stages one file / all changes
git commit -m "message"	Saves staged changes as a snapshot
git log --oneline	Shows commit history in short form
git diff	Shows unstaged changes line by line
git push	Uploads local commits to GitHub
git pull	Downloads and merges new commits from GitHub
git branch	Lists branches (add a name to create one)
git switch <branch>	Moves to another branch (git switch -c creates and moves)
git merge <branch>	Merges another branch into the current one
git remote -v	Shows the connected remote URLs

7. Branching Basics

A branch lets you try changes without touching the stable `main` branch.

```
git switch -c add-readme-details     # create and move to a new branch
# ...edit files...
git add .
git commit -m "Improve README description"
git push -u origin add-readme-details
git switch main
git merge add-readme-details
```

```
git push
git branch -d add-readme-details     # delete the finished branch
```

Alternatively, push the branch and open a Pull Request on GitHub, then click Merge pull request.

8. Clone and Pull Practice

```
# on another folder or computer
git clone https://github.com/<username>/java-internship-tasks.git

# after editing a file directly on GitHub (pencil icon), bring it down:
cd java-internship-tasks
git pull
```

9. Writing a Good README

The README is the front page of your repository. Include:

- Project title and a one-line description.
- Repository structure (folders and files).
- A table or list explaining the program in each Day folder.
- Features and the concepts each program demonstrates.
- How to compile and run (`javac` / `java`) and the required JDK version.
- Author name and GitHub link.

The accompanying `README.md` file is a ready template. Replace the Day 1-3 placeholder rows with your own topics, and put your name and username at the bottom. Use short, clear commit messages in the present tense, such as "Add Day 4 student management program".

10. Common Problems

Message / problem	Fix
fatal: not a git repository	Run git init, or cd into the repository folder.
remote origin already exists	git remote set-url origin <url>
failed to push: non-fast-forward / rejected	Run git pull --rebase, then git push. This happens when GitHub has commits you lack (for example an auto-created README).
Authentication failed	Use a Personal Access Token, browser sign-in or SSH key, not your password.
Please tell me who you are	Set user.name and user.email with git config.
Merge conflict	Open the file, keep the correct lines, delete the <<<<<<< ======= >>>>>>> markers, then git add and git commit.

11. Completion Checklist

- [ ] GitHub account created and email verified
- [ ] Git installed and name/email configured
- [ ] Repository `java-internship-tasks` created
- [ ] Day 1-4 programs placed in Day1 to Day4 folders
- [ ] README.md written and .gitignore added
- [ ] First commit pushed to the main branch
- [ ] Created, committed on and merged at least one branch
- [ ] Cloned the repo once and ran git pull
- [ ] Repository link ready to share with your mentor